

**ZooInverse**

A collaborative platform for people to partake in cutting edge astronomy research.
<http://dgit.in/mar21-09>

**Agent Exoplanet**

An initiative that allows people to help in the study of known exoplanets that orbit stars other than our Sun. <http://dgit.in/mar21-10>

A sungrazing comet captured by SOHO

Credit: ESA/NASA/SOHO

Crowdsourced Astronomy

Using the power of collective stargazing

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hen it comes to citizen science projects, there are far more in the field of astronomy than

any other. While space is exciting, telescopes and satellites gather so much data that it is impossible for scientists, their computers, and the image processing pipelines to actually process all the data. Astronomers also continuously go

over old data with new methods, going down to accounting for specific pixels and photons during the image processing pipeline, and new findings continue to be released years and even decades after the observations were made. There is just too much data waiting to be analysed, that astronomers refer to it as a “data deluge”, which is where you can help! When participating in any of these projects, it is entirely possible that you might come across something that nobody has seen before, even though the chances are rare. For example, while analysing the light curves of a star, you might come across the telltale signs of the transits in a multi planet system, or

find a comet streaking towards the Sun that you can claim as your own.

FINDING EXOPLANETS

The Kepler space telescope was purpose built to study a fixed portion of the Milky Way. For 9 years, the spacecraft stared at stars in the Cygnus, Lyra and Draco constellations. 530,506 stars were observed during the mission, and kepler checked for slight variations in the brightness of the stars every 30 minutes. The instruments on board collected more data than could be stored or beamed back to the Earth, which meant the scientists had to select the pixels associated with the stars of interest, which amounted to only 6 percent of the 95 million pixels on the 50 × 25 mm CCD sensor. The researchers then processed the raw data from the Kepler instrument, using computers to sift through the light curves of thousands of stars. By observing the variations of the stars over long periods of time, it is possible to spot times when the light coming from the star dips. If the dips are smooth and slight, they are sunspots. If they are sharp and dramatic, they are probably exoplanets moving between the star and the Earth, what is known as a “transit” event. Larger planets cause sharper dips in the brightness of the stars. While the computers were very good at going through the data and detecting exoplanets, there was a chance that the computers had missed something. For this reason, NASA released the Kepler data to the general public. Researchers from Yale then initiated a project that allowed anyone, anywhere to go through the Kepler data and identify potential exoplanet transits, by analysing the light curves themselves. If transit events are detected, they can be marked and sent to researchers for review through the app. Exo-